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Food security in Indonesia: the role of social capital

Estiana Rusmawati[a], Djoni Hartono [b,c] and Adiwan Fahlan Aritenang[d]

[a]Badan Pusat Statistik – Statistics Indonesia, Jakarta, Indonesia; [b]Faculty of Economics and Business, Universitas Indonesia, Depok, Indonesia; [c]Research Cluster on Energy Modeling and Regional Economic Analysis, Universitas Indonesia, Depok, Indonesia; [d]Urban and Regional Planning, Institut Teknologi Bandung, Bandung, Indonesia

ABSTRACT
Social capital (i.e. bonding and bridging) affects food security. In Indonesia, studies have been conducted at the site level, covering households in specific provinces, districts, subdistricts, or even villages. Accordingly, this study contributes to the existing literature by employing a sample of 68,304 households across Indonesia using data from Statistics Indonesia's (BPS's) 2018 National Socioeconomic Survey (SUSENAS). Based on Lewbel's (2012. "Using Heteroscedasticity to Identify and Estimate Mismeasured and Endogenous Regressor Models." *Journal of Business and Economic Statistics* 30 (1): 67–80) two-stage least squares (2SLS) regression test, the results suggest that bonding and bridging social capital positively affect food security. Specifically, bridging social capital has a more substantial influence on food security.

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Food security; social capital; bonding; bridging; household; Indonesia

Introduction

Food security remains an unresolved global problem. According to the *Economist Intelligence Unit*, based on the global food security index (GFSI), the average global food security in 2019 was only 62.9 of the maximum score of 100. In recent years, this condition has been exacerbated by a worldwide slowdown (and even setbacks) in food security improvement (*The Economist Intelligence Unit* 2019). Likewise, in 2019, Indonesia only scored 62.6 on its food security index, lower than the global average score. Indonesia's food security level is inferior to that of several other member states of the Association of Southeast Asian Nations (ASEAN), such as Malaysia (73.8) and Thailand (65.1) (*The Economist Intelligence Unit* 2019).

Moreover, food sufficiency is undoubtedly a basic necessity. Appropriate intake of high-quality food with adequate nutritional composition is the main prerequisite for a healthy and productive life. Food insecurity threatens social stability, security, and world peace. Thus, a low level of food security at both the global and national levels is a severe problem that urgently requires improvement (World Food Programme 2002).

The literature suggests that social capital plays a vital role in increasing household food security. Prior studies argue that social capital is beneficial for households to survive because of social and psychological support (Putnam 2000). Social cohesion and mutual trust among community members increase access to resources, external assets, job opportunities, and information dissemination by enabling people to ask for help from neighbors and relatives during challenging situations (Chriest and Niles 2018).

Scholars have conducted studies to confirm how social capital impacts food security (Dean and Sharkey 2011; Dzanja et al. 2015). Examples include Jackson et al. (2019) who performed a study in the US, Dean and Sharkey (2011) who conducted a study in the US state of Texas, and Lamidi (2019) who carried out a study in Nigeria. The findings indicate that social capital has a significant effect on food security. For instance, Dean and Sharkey (2011) found that social capital has a positive and significant impact on Texas's household food security. Low-income residents actively participate in the community, and social activities are more supportive of helping the community in challenging times. Jackson et al. (2019) stated that American households' social capital is correlated with food security. This correlation emerged through food sharing and monetary and non-monetary forms of assistance from social community members.

Research conducted in Indonesia had similar outcomes (Martianto, Alfiasari, and Dharmawan 2009; Burrahmad 2018; Nasrudin et al. 2020). However, studies are limited to sampling households in particular parts

CONTACT Djoni Hartono djoni.hartono@ui.ac.id

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of Indonesia. Several studies only cover individual provinces or districts, such as Nasrudin et al. (2020), who investigated the Maluku Islands and Burrahmad (2018), who studied Pidie Regency. Previous research also only covers particular villages, such as Martianto, Alfiasari, and Dharmawan (2009), who examined the Tanah Sareal subdistrict, Bogor and Prayitno, Baiq Maulida, and Nugraha (2019), who investigated the Ngadirejo subdistrict of Malang. This situation has impacted the generalizability of past findings to contribute to national policy development regarding food security issues.

This research contributes to the food security literature using a larger sample to obtain more representative results; the sample of 68,304 households is distributed throughout Indonesia according to the 2018 National Socioeconomic Survey (SUSENAS) conducted by Statistics Indonesia (BPS according to the abbreviation in Bahasa Indonesia). It also provides an analysis that can be employed for national policy development concerning social capital and food security in Indonesia.

Moreover, past literature on the impact of social capital on food security mostly ignores endogeneity problems that could arise from the reverse causality between social capital and food security (Putnam 2000). This study addresses the endogeneity problem using the instrumental variables (IVs) approach. More specifically, it follows Lewbel's (2012) two-stage least squares (2SLS) regression test, generating internal IVs from the auxiliary equation's residual, multiplied by each of the included exogenous variables in mean-centered form. The study used this technique due to challenges in finding truly exogenous variables, as also experienced by other scholars like Heikkilä, Kalmi, and Ruuskanen (2016) and Shoji and Murata (2021). Further, Lewbel's (2012) 2SLS approach produced a very close estimate to the one obtained from using the exogenous IVs and is therefore commonly used in prior literature, such as Acheampong, Erdiaw-Kwasie, and Abunyewah (2021), Appau, Churchill, and Farrell (2019), and Churchill and Mishra (2017).

This study is comprised of several sections. Next, the literature review discusses food security and social capital concepts and how the latter impacted the former. The section after that covers the methodology, explaining this study's data, sample, and analysis. Then the results and discussion are presented, followed by the conclusions.

Literature review

The concept of food security

Food security is a dynamic concept. The 1974 World Food Summit described food security as 'availability at all times of adequate supplies of basic foodstuffs to sustain a steady expansion of food consumption and offset fluctuations in production and prices' (FAO 1976). This definition implicitly prioritizes the 'availability' of food supplies as the primary consideration in determining a country's food security. Nonetheless, the work of Nobel Laureate Amartya Sen on poverty and famines (Sen 1981), drawing on an analysis of famines in Bengal (1943), Ethiopia (1973), and Bangladesh (1974), convincingly underlines the need to ensure everyone's access to food since there can be famine and undernourishment even when food is abundant. Accordingly, the food security definition goes beyond the availability to cover 'access' as the second food security pillar.

By the mid-1990s, alleviating micronutrient undernutrition became the primary focus of nutrition research, shifting attention from mere food sufficiency to overall diet quality. Both trends had implications for the conceptualization of household food security by involving 'utilization' as the third food security pillar (Jones et al. 2013). This pillar guarantees people's ability to use available and accessible foods appropriately and to obtain essential nutrition and health care (Fawole, Ozkan, and Ayanrinde 2016).

Later debates on food security in 1996 also revealed issues of food supply stability and having food that meets nutritional requirements for a healthy and active life. Reflecting on these developments, the definition of food security shifted from 'availability, at all times, of adequate world supplies of basic foodstuffs' in the 1970s to 'a situation that exists when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food meeting their dietary needs and food preferences for an active and healthy life' (Sseguya, Mazur, and Flora 2018). This definition added another pillar (stability), meaning that the earlier three factors (availability, accessibility, and utilization) are sustainable for a record period (Fawole, Ozkan, and Ayanrinde 2016). The food security definition adopted in 1996 is the most recent and reflects the four dimensions of the food security concept: availability, access, utilization, and stability.

Approaches to gauge this have evolved along with their concepts and definitions. These measures may focus on food availability, access, utilization, the stability of food security over time, or a combination of these domains. Carletto, Zezza, and Banerjee (2013) found that the most commonly used food security indicators involve national undernourishment, household food consumption (calorie intake), dietary diversity, the food consumption score (FCS), the Household Food Insecurity Access Scale (HFIAS), the coping strategy index, and the food adequacy question.

In theory, these measures cannot be considered on their own as a proxy for food security; previous studies have cited calorie intake as the most relevant indicator of food security (Hossain, Mullally, and Asadullah 2019). This indicator measures energy intake and can reflect both the availability and access pillars of food security. Furthermore, the higher relevance of calorie intake is easy to verify in practice; for instance, using the calorie-intake-based food security cut-off, as employed by the World Bank (2012). The report classifies households with calorie intake per capita higher than 2,100 kcals/day as being food secure. Consumption of approximately 2,100 kcal of a poorly varied diet (low dietary diversity score) daily will keep a person alive. Conversely, consuming 1,000 kcal daily with a diverse set of foods (high dietary diversity score) will not keep a person alive. Accordingly, the literature suggests calorie intake as the 'gold standard' for food security measures (Hossain, Mullally, and Asadullah 2019).

The concept of social capital

The development of the idea of social capital is inextricable from the contributions of three prominent figures: Pierre Bourdieu, James Coleman, and Robert Putnam. Bourdieu (1986) defined social capital at the micro level. He suggested that social capital entails individual access to resources like information, influence, opportunities, and financial assistance. This view has been a central element of the social capital literature. Coleman (1988) observed social capital from a broader, more optimistic angle involving micro – and meso-level networks. He examined the role of family interactions in individual outcomes such as education level, and the role of public relations, norms, and sanctions in group outcomes.

On the other hand, Putnam (2000) focused on social capital operating at society's macro and meso levels. He interpreted social capital in terms of reciprocity norms and civic engagement networks, and highlighted trust as a critical aspect of social capital. He explained that social networks mix 'horizontal' and 'vertical' relationships. Horizontal relationships unite agents with equal status and power, whereas vertical ones link unequal agents in asymmetric hierarchical and dependent interactions.

Following these three authors' lines of thinking, the concept of social capital is still developing without absolute agreement on its definition and measurement (Scrivens and Smith 2013). However, there is a general tendency to define social capital as the groups, networks, norms, and beliefs held by society for productive purposes. Scholars agree that social capital is a multidimensional notion, and scholars have explored it at various levels, i.e. micro (individual), mezzo (community), and macro (national or regional) (Sseguya, Mazur, and Flora 2018).

Based on its manifestation of a multidimensional and multilevel concept, Flora, Flora, and Gasteyer (2016) divided social capital into bonding and bridging. Bonding social capital represents a relationship between people of the same ethnicity, social status, and location. Social cohesion refers to social cohesion within a group or community rooted in shared beliefs and moral values strengthened by a cooperative bond (Narayan and Pritchett 1999). Examples of bonding social capital include social cohesion at the local/small scale, such as family ties, kinship, religion, and non-governmental organizations (NGOs). Theoretically, the connection tying these groups together, coupled with a common background, drives reciprocity and solidarity within society (Hofferth and Iceland 1998; Helliwell and Putnam 1999).

Bridging social capital involves relationships and networking across social groups. This notion includes coordination, collaboration with other groups, and social support or information sharing across communities (Narayan and Pritchett 1999). Bridging social capital contains an economic relationship, and legal and formal institutions frequently involve cultural and regional differences. Moreover, bridging social capital connects various household, environmental, government, and non-governmental groups. It also incorporates people into interlevel communities, such as those with different economic statuses (Helliwell and Putnam 1999). Bridging social capital can be crucial in collective resource organizations without government interaction (Chriest and Niles 2018). This type of social capital provides information sharing and models to assist community members.

The correlation between social capital and food security

Previous literature suggests that social capital could play a role in the form of norms and networks as a precondition for economic growth (Putnam 1993). Social networks enable communication and coordination, fostering trust among community members. This mutual trust strengthens the norms of necessity for people to help each other. Ultimately, various benefits achieved through past cooperation will promote ongoing collaboration in the future. Leonard (2004) argued that social capital is the glue that bonds the benefits of economic and physical capital in marginalized communities. He expressed that social capital can be converted into other forms of capital, and can

encourage new investments and make existing ones go further. Accordingly, Putnam (2000) asserted that social capital is essential for low-income people with little economic, physical, or human capital.

Nosratabadi et al. (2020) suggested that social capital improves food security through its pillars: food availability, food accessibility, food utilization, and food system stability. Practically, this improvement occurs through knowledge and product sharing. An example of the knowledge-sharing process mentioned by Malual and Mazur (2022) is that kinship and friendship ties operate as informal information channels for job opportunities, mainly through word of mouth. Hence, family and community ties are vital for individuals to earn adequate income to meet their daily needs.

Furthermore, scholars explained that product-sharing mechanisms could take the form of the reception of food/money, loans, and access to lower-priced products through networks and communities. The latter is possible because low-income populations encourage the growth of informal economic activities by increasing the demand for low-cost goods and services. To meet this demand, small-scale retailers operate outside of recommended safety and health standards. These retailers can supply low-income communities with cheaper goods and services because they work without adhering to state legislations (Malual and Mazur 2022). Agnitsch, Flora, and Ryan (2006) posited that social capital increases the efficiency of other forms of capital because a group with greater trust will produce more efficiently than that with lower social capital.

Consequently, it is undeniable that social capital contributes significantly to increasing households' food security. Putnam (2000) maintained that this notion is valid for bonding and bridging social capital. Bonding social capital helps community members survive through close support among themselves (i.e. *getting by*). Meanwhile, bridging social capital enables better household access to resources by providing opportunities to move forward (i.e. *getting ahead*) through cross-community links involving people from various backgrounds (Grootaert et al. 2004). In line with this notion, Putnam (2000) also suggested that bonding social capital benefits household survival because of social and psychological support. This support is crucial for community members in poor conditions, including food insecurity.

For instance, it allows households to obtain loans and job opportunities from local entrepreneurs. Bridging social capital offers community members greater potential to escape challenging conditions. It provides links to external assets and disseminates information. Additionally, it connects people to a more diverse environment and is lucrative in terms of finding a job because of broader opportunities. Higher bonding and bridging social capital can encourage food-insecure individuals to borrow foodstuffs or money and more easily access other resources to meet their food-related needs. This high level of social capital also urges individuals to absorb more information and to find role models to improve their food security (Chriest and Niles 2018).

Previous studies on social capital and food security

Studies have examined the role of social capital in boosting household food security. Dean and Sharkey (2011) concluded that households with low social capital in Texas tend to experience food insecurity. Past research indicates that social capital is essential for reducing food insecurity, increasing fruit and vegetable intake, and reducing the risk of hunger and childhood malnutrition. Likewise, Sseguya, Mazur, and Flora (2018) surveyed 378 households in rural Uganda and found that families with bridging and linking social capital, characterized by membership in groups, access to information from external institutions, and observance of norms in groups tended to be more food secure.

Chriest and Niles (2018) studied the role of bonding and bridging social capital in enhancing food security. They examined the food security levels of rural communities in the US state of Vermont during extreme weather events (EWEs) in 2011. They found that those with greater social capital before the EWEs effectively improved and maintained food security. However, the opposite condition did not negatively affect food security, provided that the community could build bridging social capital during and after the EWEs. Thus, they confirmed that rural communities with high bonding and bridging social capital can effectively overcome short-term food insecurity during and after EWEs.

Lamidi (2019) explained that social networks and informal communities benefit underprivileged households by providing monetary and non-monetary assistance. This finding implies that support from community social networks, such as unsecured loans, can significantly diminish food insecurity. Finally, Malual and Mazur (2022) studied the associations between social capital and food security in Lira District in northern Uganda, focusing on the impact of a post-conflict scenario. They also found a strong positive association between social capital and household food security.

Research methodology

Data

This study used data from BPS's 2018 SUSENAS. BPS conducted the survey in September 2018 with households as the analysis unit. The survey involved 68,304 households distributed across 34 Indonesian provinces and 514 regencies/cities. BPS conducted a unit sample selection using a two-stage stratified sampling method. The first stage selected 7,500 blocks from 30,000 census blocks estimated by district/city based on predetermined urban/rural welfare stratum allocations in each district or city. Next, households were selected from the updated blocks using a systematic sampling method with implicit stratification based on the households' welfare level. BPS used the household head's highest education level as a proxy for households' welfare level (BPS 2018).

Operational definitions of the variables

This section presents operational definitions of the variables (the dependent, independent, and control variables). Table A1 in the Appendix summarizes their operational definitions.

Dependent variable

The dependent variable was household diet quality, and food security was measured using a food consumption indicator or calorie intake (*calorie intake*). Calorie intake is the number of calories that households consume from their food composition (Fawole, Ozkan, and Ayanrinde 2016). As mentioned in the literature review, calorie intake is the 'gold standard' of food security indicators; it can reflect the availability and access pillars, posing a higher relevance to the notion of food security.

This study obtained calorie intake data from the database of BPS's 2018 SUSENAS. The data show the daily household calorie intake per capita during the seven days before the survey. The present study employed per capita calorie intake data to make a fair comparison between families of different sizes or household members. The use of this measure for food security-related studies is not rare in prior literature, such as Maharjan and Khatri-Chhetri (2006), Tefera and Tefera (2014), Muleta, Ketema, and Ahmed (2021), and Oskorouchi and Sousa-Poza (2021).

Independent variables (The variables of interest)

The variables of interest were households' social capital, classified as bonding and bridging. It used data on households' participation in neighborhood activities – such as religious events, skills development, sports/games, community services, social gatherings, funeral ceremonies, family gatherings, art shows, and children's health-monitoring services – to measure bonding social capital (*bonding*). The measurement of bridging social capital (*bridging*) involved data on households' participation in organizations outside of people's personal lives, schools, or work environments. Other measures employed for this variable include households' tolerance toward other ethnic groups and religious activities (Zhang, Anderson, and Zhan 2011). Table A3 in the Appendix presents SUSENAS' questions for the *bonding* and *bridging* social capital index measurements.

In addition, the data used to gauge bonding and bridging social capital are found in BPS's 2018 SUSENAS, measured in the form of 'Yes' or 'No' answers. Filmer and Pritchett (2001) recommended using principal component analysis (PCA) to develop socioeconomic variables to avoid subjectivity. Based on the answers, the present study employed PCA to generate indices to determine bonding and bridging social capital. PCA is a factor analysis technique that is utilized to simplify the correlation matrix between variables. PCA is superior to other kinds of factor analysis because it enables each variable's main component to explain the variation in the correlations among the variables themselves (Ariawan 2006).

This study followed Kolenikov and Angeles (2004) in utilizing a tetrachoric correlation matrix for PCA. Kolenikov and Angeles (2004) argued that this approach is more suitable for categorical variables. The tetrachoric correlation matrix for PCA addresses the spurious correlation between the dummy variables representing the answers by employing ordinary PCA for nominal data, such as items that can be answered with 'Yes' or 'No.'

The control variables

This study involved several control variables to avoid omitted variable bias. Furthermore, based on past literature, we employed household and household head characteristics as the control variables.

Lamidi (2019) found an inverted U-shaped relationship between household members and food security: households with more members are more food secure until a certain level, where a higher number of family members results in more severe food insecurity. Moreover, Asa and Archibong (2016) and De Marco and Thorburn (2009) confirmed that food insecurity is common among underprivileged families. Drewnowski and Specter (2004) showed that lower household income is associated with a higher possibility of severe food insecurity. However, Sundari and Nachrowi (2015) reported that those benefitting from food subsidy programs

have better food security levels. Finally, Dean and Sharkey (2011) verified that families in rural areas have a greater possibility of experiencing food insecurity. Likewise, Mahadevan and Suardi (2014) suggested that households living in a more progressive state had a greater opportunity to effectively improve their food security due to the difference in infrastructure quality between states.

Yamin and Dartanto (2016) found an inverted U-shaped association between the household head's age and welfare, including food security. The older the household head, the more food secure the household is until a certain level, where the opposite occurs. Moreover, Lamidi (2019) indicated that households with less educated or unemployed heads are less food secure. Likewise, Dean and Sharkey (2011) and Asa and Archibong (2016) concluded that female-headed households experience more severe food insecurity. Finally, Martin et al. (2004) suggested that widowed-headed households experience more severe food insecurity than others.

Consequently, the household characteristics (control variables) in this study include the number of household members (*HHsize*) and household members squared ($HHsize^2$), poverty status (*Dummy_poor*), the Raskin recipient status (*Dummy_yesRaskin*), and spatial status (*Dummy_urban*). On the other hand, the household head's characteristics (control variables) are age (*Age*), age squared (Age^2), education level (*Education*), gender (*Dummy _male*), marital status (*Dummy _widowed*), and employment status (*Dummy_employed*).

Data analysis

To determine whether bonding and bridging social capital are associated with food security, the present study conducted a regression analysis based on the following specification models:

$$Calorie_intake_i = \alpha_0 + \alpha_1 \, Bonding + \gamma X_i + \varepsilon_i \quad (1)$$

$$Calorie_intake_i = \beta_0 + +\beta_1 \, Bridging + \delta X_i + \theta_i \quad (2)$$

Coefficient α_1 denotes the impact of household bonding social capital on food security, measured using calorie intake. Meanwhile, β_1 gauges the impact of bridging social capital on household food security. Finally, X_i represents the control variables.

Empirical strategy

This study acknowledges the endogeneity problem arising from the reverse causality between social capital and food security. Meanwhile, previous empirical evidence indicates that social capital affects food security (Dean and Sharkey 2011; Lamidi 2019; Sseguya, Mazur, and Flora 2018). On the other hand, a low level of food security can lead to low levels of social capital, as Putnam (2000) demonstrated.

The literature proposes an alternative solution to the endogeneity problem by including IVs in the regression model. More specifically, Acheampong, Erdiaw-Kwasie, and Abunyewah (2021) explained that an IV (z) of regression y on x is a variable meeting three criteria: (1) z does not have a direct effect on y; (2) z is not associated with the regression error term; and (3) z strongly associates with x. However, Sila, Gonzalez, and Hagendorff (2016) argued that finding a genuinely exogenous IV can be challenging for an empirical study.

Likewise, the study cannot find an IV estimator that explains food security but is exogenous to bonding and bridging social capital. This situation has occurred in other studies such as Heikkilä, Kalmi, and Ruuskanen (2016) and Shoji and Murata (2021). Accordingly, following Acheampong, Erdiaw-Kwasie, and Abunyewah (2021), this study employed Lewbel's (2012) 2SLS regression test, applied when an exogenous IV is unavailable or weak.

Lewbel's (2012) 2SLS approach is crucial to address the issue of endogeneity because it generates internal IVs from the residual of the auxiliary equation, multiplied by each of the included exogenous variables in mean-centered form. Lewbel's (2012) 2SLS approach produced a very close estimate to that obtained using exogenous IVs. Moreover, previous studies have commonly applied this technique, such as Acheampong, Erdiaw-Kwasie, and Abunyewah (2021), Appau, Churchill, and Farrell (2019), and Churchill and Mishra (2017). Applying Lewbel's (2012) 2SLS method could effectively address the endogeneity problem identified in the regression models. Finally, employing Lewbel's (2012) 2SLS approach and referring to the literature review on the effect of social capital on food security, this study proposes the following hypotheses:

H1a: Bonding social capital positively affects household food security

H1b: Bridging social capital positively affects household food security.

Results and discussion

Descriptive statistics

Table 1 summarizes the descriptive statistics of the variables involved in this study.

We can see that a household member, on average, consumes 2,216.36 kcals per day. Consequently, based on the World Bank (2012) food security cut-off, the

Table 1. Summary Statistics of the Variables.

Variables	Obs.	Mean	Std. Dev.	Min	Max
Calorie_intake	68,304	2,216.36	583.63	1,235.66	3,690.84
Bonding	68,304	1.39	0.75	0	2.91
Bridging	68,304	0.40	0.41	0	1.84
HHSize	68,304	3.97	1.54	2	18
$HHSize^2$	68,304	18.16	15.47	4	324
Dummy_poor	68,304	0.11	0.31	0	1
Dummy_yesRaskin	68,304	0.27	0.44	0	1
Dummy_urban	68,304	0.43	0.50	0	1
Age	68,304	48.14	12.82	18	97
Age^2	68,304	2,482.69	1,310.45	324	9,409
Education	68,304	0.52	0.66	0	2
Dummy_male	68,304	0.88	0.32	0	1
Dummy_widowed	68,304	0.12	0.33	0	1
Dummy_employed	68,304	0.90	0.29	0	1

study samples were generally food secure. Nonetheless, the minimum value of 1,235.66 kcals per day infers that a non-food-secure group still exists in Indonesia. Furthermore, the mean of the variables of interest was 1.39 for *bonding* and 0.47 for *bridging*, respectively. This situation is in line with Woolcock and Narayan (2000) and Flora and Thiboumery (2005), who stated that *bonding* is relatively higher than *bridging* in middle-lower income countries.

Estimation results

Four ordinary least squares (OLS) regressions were run to test the hypotheses, as shown in Table 2.

As illustrated in columns I and III of Table 2, the study initially regressed the dependent variable (*calorie intake*) on social capital alone (*bonding* and *bridging*). As presented in columns II and IV, the model involved control variables (household and household head characteristics). These tests indicate a significant association between social capital (*bonding* and *bridging*) and food security (*diet diversity*) at the 1% significance level. However, this study acknowledges a potential endogeneity problem in its model arising from the reverse causality between food security and social capital.

To address the endogeneity problem that may arise from the reverse causality between food security and social capital, Lewbel's (2012) 2SLS approach was employed (see Table 2). The table indicates that *bonding* and *bridging* had a statistically significant impact on *food security*. Furthermore, the coefficient of *bridging* (130.1) was considerably higher than the coefficient of *bonding* (83.69).

Robustness check

To check the robustness of the estimation models, this study conducted sensitivity analyses by alternating food security measures. It replaces its food consumption indicator (*calorie_intake*) with dietary diversity measures following Hossain, Mullally, and Asadullah (2019), who suggested dietary diversity as a commonly used proxy for calorie intake. They also reported that dietary diversity gained prominence owing to its close relationship with household access to food and calorie availability.

More specifically, it employed the most commonly used dietary diversity measures suggested by Hatløy et al. (2000): the dietary diversity score (DDS) and the food variety score (FVS). This study generated the DDS and FVS from the database of BPS's 2018 SUSENAS.[1] DDS is the number of food groups in the diet consumed by each household during the seven days prior to the survey. On the other hand, FVS is the number of food items consumed by any household during the seven days before the survey (Hatløy et al. 2000; Headey and Ecker 2013; Fawole, Ozkan, and Ayanrinde 2016). Table 3. outlines the robustness test results for DDS and FVS.

As seen in the table, bonding and bridging statistically and positively affect food security for the DDS and FVS measures. Moreover, the coefficients of bridging (DDS: 0.348, FVS: 5.734) were significantly higher than those of bonding (DDS: 0.144, FVS: 1.092). These results confirm the robustness of the two SLS models listed in Table 2.

Discussion

This study found a positive and statistically significant effect of bonding and bridging social capital on household food security (*calorie_intake*). These findings remained robust after replacing *calorie_intake* with DDS and FVS. In other words, this study confirmed that bonding and bridging social capital play crucial roles in dealing with food shortages through brotherhood networks, friendships, and close ties with neighbors. These outcomes align with those of Chriest and Niles

Table 2. Ordinary Least Squares (OLS) Regression and Two-Stage Least Squares (2SLS) Estimation Results.

	OLS Model				Lewbel (2012) 2SLS	
Dependent Variable: *Calorie_intake*	I	II	III	IV	Model 1	Model 2
Bonding	45.41***	25.49***			83.69***	
	(15.44)	(9.38)			(3.67)	
Bridging			119.9***	78.83***		130.1***
			(22.31)	(13.43)		(13.86)
HHsize		−271.3***		−270.7***	−272.8***	−270.8***
		(−56.17)		(−56.10)	(−55.90)	(−56.09)
HHsize^2		15.66***		15.59***	15.88***	15.61***
		(32.97)		(32.86)	(32.79)	(32.89)
Dummy_poor		−432.0***		−427.2***	−425.9***	−422.3***
		(−64.92)		(−64.07)	(−60.11)	(−62.97)
Dummy_yesRaskin		−5.047		4.807	−8.796	10.15*
		(−1.09)		(1.03)	(−1.80)	(2.14)
Dummy_urban		38.75***		39.02***	35.14***	38.17***
		(9.60)		(9.68)	(8.19)	(9.46)
Age		26.23***		28.21***	23.34***	28.68***
		(25.61)		(27.77)	(15.33)	(28.16)
Age^2		−0.234***		−0.248***	−0.207***	−0.250***
		(−23.01)		(−24.65)	(−14.18)	(−24.82)
Education						
Secondary_School		12.44**		1.173	7.915	−7.433
		(2.69)		(0.25)	(1.60)	(−1.53)
Higher_Education		84.15***		34.53***	75.31***	−0.267
		(11.40)		(4.12)	(9.22)	(−0.03)
Dummy_male		33.01**		28.24**	44.38***	28.39**
		(3.29)		(2.82)	(4.03)	(5.45)
Dummy_widowed		−47.12***		−48.25***	−40.89***	−47.20***
		(−4.81)		(−4.93)	(−4.04)	(−4.82)
Dummy_employed		61.19***		66.31***	44.88***	64.99***
		(7.87)		(8.58)	(4.47)	(8.40)
Constant	2,153.5***	2,258.8***	2,168.2***	2,203.3***	2,269.3***	2,170.3
	(463.23)	(84.12)	(698.64)	(81.31)	(83.28)	(13.86)
Adj R^2	0.0035	0,2044	0.0072	0.2055		
Cragg-Donald F Test					82.58	3,662
Observations	68,304	68,304	68,304	68,304	68,304	68,304

*, **, and *** indicate 10%, 5%, and 1% significance level.

(2018), who indicated that communities with high social capital can effectively improve and maintain food security stability. Yamin and Dartanto (2016) revealed that bonding and bridging social capital strengthen households' ability to escape poverty.

This study also revealed an interesting finding by showing a stronger impact of bridging social capital on household food security. Table 2 demonstrates that the coefficient of bridging (130.1) was significantly higher than the coefficient of bonding (83.69). This situation is due to the differences in the characteristics of bonding and bridging in supporting households facing difficulties. Bonding social capital plays a more crucial role in surviving challenging conditions (*getting by*), while bridging plays a more significant role in enabling people to overcome challenges (*getting ahead*) (Grootaert et al. 2004). Bridging social capital represents interactions among people with diverse backgrounds, enabling them to overcome difficulties by getting information, learning processes, and understanding others' experiences. In addition, economic strata variations can increase food-insecure people's opportunities to obtain monetary and non-monetary assistance from their bridging communities (Coffé and Geys 2007; Chriest and Niles 2018; Lamidi 2019). This finding suggests the importance of bridging the gap between different socioeconomic groups. Accordingly, bridging social capital will have a higher impact on household food security (Chriest and Niles 2018; Sseguya, Mazur, and Flora 2018).

Ironically, empirical evidence implies that constructing bridging social capital is more complicated than developing bonding social capital. It is more difficult for community groups to become involved in heterogeneous community networks. Social barriers, such as differences in demographic and sociocultural traits (ethnicity, religion, social status), prevent them from interacting with diverse household networks (The World Bank 2001). Thus, fostering bridging social capital within these communities requires prominent leadership activities such as community, cultural, religious, and local government leaders (Yamin and Dartanto 2016).

Based on these findings, this study underscores the importance of current policies promoting food diversity and security in Indonesia, as implemented by the Food Security Agency (Badan Ketahanan Pangan, or BKP), which is overseen by the Ministry of Agriculture.

Table 3. Two-Stage Least Squares (2SLS) Estimation Results Involving DDS and FVS.

	Dependent Variable: DDS		Dependent Variable: FVS	
Independent Variable	(1)	(2)	(1)	(2)
Bonding	0.144*		1.092*	
	(2.21)		(2.39)	
Bridging		0.348***		5.734***
		(12.87)		(30.58)
HHsize	0.459***	0.463***	5.061***	5.081***
	(32.84)	(33.23)	(51.73)	(52.67)
HHsize^2	−0.0336***	−0.0340***	−0.338***	−0.340***
	(−24.17)	(−24.83)	(−34.83)	(−35.86)
Dummy_poor	−0.792***	−0.774***	−7.525***	−7.094***
	(−38.98)	(−40.01)	(−52.98)	(−52.94)
Dummy_yesRaskin	−0.257***	−0.212	−2.753***	−2.085***
	(−18.36)	(−7.51)	(−28.09)	(−21.98)
Dummy_urban	0.105***	0.108***	1.658***	1.630***
	(8.55)	(9.30)	(19.28)	(20.21)
Age	0.0349***	0.0452***	0.347***	28.68***
	(7.99)	(15.39)	(11.35)	(28.16)
Age^2	−0.0003***	−0.0004***	−0.0031***	−0.250***
	(−7.08)	(−12.91)	(−10.47)	(−24.82)
Education				
Secondary_School	0.241***	0.194***	1.555***	−7.433
	(16.99)	(13.80)	(15.95)	(−1.53)
Higher_Education	0.322***	0.108***	0.601**	−0.267
	(13.77)	(3.84)	(3.09)	(−0.03)
Dummy_male	0.185***	0.158***	0.886***	28.39**
	(5.86)	(5.45)	(4.02)	(5.45)
Dummy_widowed	−0.204***	−0.212***	−1.172***	−47.20***
	(−7.02)	(−7.51)	(−5.78)	(−4.82)
Dummy_employed	0.0138	0.0452***	1.002***	64.99***
	(0.48)	(15.39)	(4.97)	(8.40)
Constant	9.013***	8.762***	16.75***	2,170.3
	(115.40)	(110.44)	(30.66)	(13.86)
Cragg-Donald F Test	82.58	3,662	82.58	3,662
Observations	68,304	68,304	68,304	68,304

*, **, and *** indicate the 10%, 5%, and 1% significance levels, respectively.

Programs offered include Sustainable Yard (Pekarangan Lestari, or P2L), Farming Corporation Development (Pengembangan Korporasi Usahatani, or PKU), Community Food Granary (Lumbung Pangan Masyarakat, or LPM), Family Farming (Pertanian Keluarga, or PK), and the Indonesian Farmers' Shop (Toko Tani Indonesia, or TTI). Through these programs, Indonesia's government, particularly the BKP, helps potentially food-insecure people via community formation, such as through farming groups (BKP 2020).

In this study, the communities cultivated by the BKP can lead to the development of bonding social capital among society members with the same characteristics. However, this study and previous empirical research indicate that the effect of bonding social capital on household food security is somewhat lower than that of bridging social capital. Hence, this study recommends that more heterogeneous community members be involved in the program to foster effective household food security.

For instance, the government can engage entrepreneurs in financing programs through corporate sustainability actions. The government can also help experts in related fields by providing periodic guidance and supervision. The government can assist potential consumers in expanding their marketing networks and directly input the community's products. The involvement of various parties with heterogeneous backgrounds facilitates bridging social capital within communities. As suggested by this study and previous literature, this situation will lead to resource and information sharing, resulting in the program's success and, finally, improved household food security.

Furthermore, concerning the control variables, the impact of the number of household members (*HHsize*)on food security produced an inverted U-shape for *calorie_intake* and a parabolic shape for *DDS* and *FVS*. A deeper analysis revealed that the inflection point was at household members, equal to 9 for *calorie_intake* and 7 for *DDS* and FVS. This means that for the *calorie_intake* estimation models, increases in *HHsize* decrease *calorie_intake* to some extent until *HHsize* is more than 9; the rise in the figure will lead to an increase in *calorie_intake*. Meanwhile, for *DDS* and *FVS*, the statistics increase with more household members until they decline once *HHsize* is greater than 7.

Based on further checks of the data related to *HHsize*, *calorie_intake, DDS,* and *FVS*, there are at least two possible explanations for this inconsistency. First, this study utilized calorie intake per capita and household-level *DDS* and *FVS*. Under these circumstances, the rise in *HHsize* tends to increase household diet variety. On the other hand, despite the increase in variation, more household members will undoubtedly reduce calorie intake per capita.

The second alternative approach to dealing with the inconsistency refers to *HHsize* before and after the inflection points. For the *calorie_intake* estimation model, the sample of households with more than nine members is just 381, or approximately 0.6% of all observations. Meanwhile, for *DDS* and *FVS*, the number of families with more than seven members was approximately 2.7% of the total sample; that is, 1,820. These numbers are insignificant, and in the case of inconsistencies, Wooldridge (2016) suggests ignoring observations. In other words, the study found a negative effect for the *calorie_intake* model and a positive effect for the *DDS* and *FVS* models. This negative effect aligns with Tefera and Tefera (2014) and Muleta, Ketema, and Ahmed (2021), whereas the positive impact is in line with Grootaert et al. (2004) and Hassan and Birungi (2011).

Several control variables, namely *Dummy_urban, Dummy_male*, *Dummy_education,* and *Dummy_employed,* affect food security positively. In other words, these characteristics can prevent households from experiencing food insecurity, as indicated by previous studies such as Dean and Sharkey (2011), Mahadevan and Suardi (2014), and Lamidi (2019). Furthermore, confirming Yamin and Dartanto (2016), this study reports a parabolic impact of the household heads' age on food security. The inflection points of the study estimations are between 56 and 60 years old,[2] with those involving bridging resulting in higher figures. Finally, the remaining control variables, *Dummy_poor, Dummy_yesRaskin,* and *Dummy_widowed* lead to food insecurity. These outcomes support those of Dean and Sharkey (2011), Sundari and Nachrowi (2015), and Asa and Archibong (2016).

Conclusions and policy recommendations

Food security remains an unresolved global and national issue in Indonesia. Numerous studies have been conducted to improve related conditions. One field of study examined the role of social capital in increasing household food security. However, these studies are still not thorough enough in covering Indonesia since they only examined households in a particular area such as a province, district, subdistrict, or village. This study contributes to the existing literature by employing a more comprehensive and representative sample, enabling the generalization of the findings to serve as a basis for national policy recommendations on Indonesia's food security issues.

This study examined the effect of bonding and bridging social capital on household food security using Lewbel's (2012) 2SLS regression. This approach was used due to the reverse causality between social capital (bonding and bridging) and food security, leading to endogeneity problems. IVs were used to address this issue.

Lewbel's (2012) 2SLS regression results imply that bonding and bridging social capital positively and statistically affect household food security. Furthermore, the effect of bridging social capital on household food security was significantly stronger than that of bonding. Bonding social capital plays a more crucial role in surviving challenging conditions (*getting by*), while bridging plays a more significant role in enabling people to overcome challenges (*getting ahead*). Bridging social capital represents interactions among people of diverse backgrounds, allowing them to overcome difficulties by getting information, learning processes, and understanding others' experiences. In addition, economic strata variations can increase food-insecure people's opportunities to obtain monetary and non-monetary assistance from their bridging communities. Thus, bridging social capital has a greater impact on household food consumption.

The Indonesian government has continuously implemented policies to address the food insecurity problem, aiming to enable food-insecure households to independently produce their food through local community formations, such as farming groups. These communities, consisting of people with a homogenous background, result in bonding social capital. However, this study and past empirical evidence highlight a weaker effect of bonding social capital on household food security. This study recommends incorporating bridging social capital among communities by involving more heterogeneous participants, such as entrepreneurs, farming experts, and potential consumers, in implementing policies to address food insecurity.

Notes

1. Table A2 in the Appendix presents the food group list provided by SUSENAS 2018; we utilised it to develop our food security variable using the dietary diversity score (DDS and FVS).
2. The inflection points for *calorie_intake* estimation were 56 (bonding) and 57 (bridging); for *DDS* they were 59 (bonding) and 60 (bridging), and for *FVS* they were 57 (bonding) and 60 (bridging).

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Djoni Hartono http://orcid.org/0000-0002-9413-1903

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Appendix

Table A1. Operational Definitions of the Variables.

Variable Name	Operational Definition
Calorie_intake	Per capita daily calorie intake data during the last seven days of BPS's 2018 SUSENAS
DDS	The number of *food groups* (maximum: 13) consumed by any household during the past seven days before the survey
FVS	The number of *food items* (maximum: 188) consumed by any household during the past seven days before the survey
Bonding	Household head's bonding social capital
Bridging	Household head's bridging social capital
HHSize	The number of members in the household
Dummy_poor	Household poverty status, coded as 1 if per capita income is lower than the provincial poverty line and 0 if otherwise
Dummy_yesRaskin	Household food subsidy status, coded as 1 if a food subsidy recipient and 0 if otherwise

(*Continued*)

Table A1. Continued.

Variable Name	Operational Definition
Dummy_urban	Household spatial status, coded as 1 if living in an urban area and 0 if otherwise
Age	Household head's age based on his/her last birthday
Education	Household head's education level, coded as 0 for elementary school or lower, 1 for junior high school, 2 for senior high school, 3 for an undergraduate degree, and 4 for a postgraduate degree
Dummy_male	Household head's gender, coded as 1 if male and 0 if female
Dummy_widowed	Household head's marital state, coded as 1 for divorced or widowed-head household and 0 if otherwise
Dummy_employed	Household head's employment status, coded as 1 for employed and 0 if otherwise

Table A2. Dietary Diversity Score (DDS) Food Groups.

No.	Food Group	No.	Food Group
1	Grains	8	Fruits
2	Tubers	9	Oil and coconuts
3	Fish/Shrimp/Shellfish	10	Beverages
4	Meat	11	Seasonings
5	Egg/Milk	12	Other consumption
6	Nuts	13	Prepared food
7	Vegetables		

Table A3. Question for Bonding and Bridging Social Capital Index Measurement.

No.	Bonding Social Capital	No.	Bridging Social Capital
1	Religious events	1	Participation in organizations outside people's daily lives, school, or work environments
2	Skills development	2	Tolerance toward other ethnic groups' activities
3	Sports/games	3	Tolerance toward other religions' activities
4	Community services		
5	Social gatherings		
6	Funeral ceremonies		
7	Family gatherings		
8	Art shows		
9	Children's health monitoring service		